USN | | | | | | | | | | [22AS31]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER - 2023

INTRODUCTION TO AEROSPACE VEHICLES AND SYSTEMS

Time: 3 Hours Maximum **Marks: 100**

Instructions:
- **i.** Missing data may be suitably assumed.
- **ii.** Answer all questions

		PART - A			**5 X 2=10**
1.	**a.**	Define the term airfoil.	1	1	**02**
	b.	What is the thickness in NACA 2412 airfoil? **i.** 12% **ii.** 24% **iii.** 41% **iv.** 2% Explain the nomenclature of this airfoil	2	2	**02**
	c.	With illustration explain the various types of wing configurations.	1	3	**02**
	d.	Name the landing gear system used in old piston engine aircrafts.	1	4	**02**
	e.	State the principle and various types of rocket propulsion.	1	5	**02**
		PART - B			**6 X 5=30**
2.	**a.**	Outline the term ISA with respect to aviation and how is it useful in the aviation field.	2	1	**05**
		OR			
	b.	Derive an equation of standard atmosphere taking lapse rate into consideration.	3	1	**05**
3.	**a.**	Explain the swash plate mechanism of a helicopter with neat illustrations.	2	2	**05**
		OR			
	b.	With a basic cross-section of an airfoil and explain **i.** Chord **ii.** Mean Camber Line **iii.** The Angle of Attack **iv.** Symmetrical Airfoil **v.** Lift Coefficient	2	2	**05**
4.	**a.**	Write the characteristics of materials used in the aircraft construction. Define the terms. **i.** Ductility **ii.** Malleability **iii.** Creep **iv.** Brittleness	2	3	**05**
		OR			
	b.	Enumerate the various types of air-breathing engines.	2	3	**05**
5.	**a.**	Write a short note about the pneumatic systems in aircraft.	2	4	**05**
		OR			
	b.	Illustrate the fuel systems in aircraft with neat diagrams.	2	4	**05**

6.	a.	Briefly describe the importance of V-n diagram with neat illustrations.	2	5	**05**
		OR			
	b.	Explain in detail the various types of wing forms.	2	5	**05**
7.	a.	Write short notes on various types of missiles.	2	6	**05**
		OR			
	b.	Sketch and explain the typical pressure feed system in liquid propellant rockets.	2	6	**05**
		PART - C		**6 X 10=60**	
8.	a.	Explain in detail the contribution of the Wright brothers to aviation history.	2	1	**10**
		OR			
	b.	Calculate the standard atmosphere values of T, p, and ρ at a geopotential altitude of 14 km.	2	1	**10**
9.	a.	Explain the flight controls of a helicopter and illustrate them with respect to **i.** Forward flight **ii.** Vertical flight **iii.** Helicopter heading	2	2	**10**
		OR			
	b.	Draw and explain the wing construction of a basic aircraft and describe the components.	2	2	**10**
10.	a.	Explain in detail the working principle and cycle of a turbofan engine and interpret the category of aerospace propulsion engines. Create the appropriate simple line diagrams.	2	3	**10**
		OR			
	b.	Explain the working principle of the Ramjet engine.	2	3	**10**
11.	a.	Explain with neat diagrams the typical fuel system construction in aircraft.	3	4	**10**
		OR			
	b.	Explain with neat diagrams the typical aircraft instruments with detailed diagrams.	3	4	**10**
12.	a.	Illustrate and explain the typical landing gear construction in aircraft.	2	5	**10**
		OR			
	b.	Illustrate and explain typical flight control system using various control surfaces.	2	5	**10**
13.	a.	Describe the principle and workings of a hybrid propellant rocket engine with a neat sketch.	2	6	**10**
		OR			
	b.	Write short notes on staging and stage separation in rockets.	2	6	**10**